FUNDAMENTALS OF AI

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What Is AI?

Learn what artificial intelligence really means, where it came from, and how it fits with machine learning.

What AI means, and what it does not

You hear the words "artificial intelligence" every day. News, phones, apps and ads all use them. But the words are used so loosely that they can mean almost anything. This section gives you a clear, simple meaning you can rely on.

A working definition

The OECD is a group of countries that writes shared rules and guidance on technology. It defines an **AI system** as a machine-based system that works out, from the input it receives, how to produce an output. The output can be a prediction, new content, a recommendation or a decision. It is built to reach goals that humans set.

In simple words, an AI system does three things:

1. Takes input

Data comes in. It can be a photo, a sentence, a voice clip, or numbers from a sensor.

2. Works out an answer

The system does not follow one fixed script for every case. It infers, which means it works out the answer from what it received.

3. Gives an output

A prediction, a recommendation, a decision, or new content such as text or an image.

A spam filter fits this. It takes an email as input. It works out whether the email looks like spam. It gives a decision: send it to the spam folder, or keep it in the inbox.

> **Good to know:** AI systems differ in how independently they act. Some keep learning and adjusting after they are put to use, such as a video app that adapts to what you watch.

What AI is not

- **It is not a human mind.** IBM describes AI as technology that lets machines *simulate* human learning and similar abilities. Simulating something is not the same as being it.
- **It is not one single thing.** A route suggestion in a maps app, a face-grouping feature in a photo app and a chatbot are all AI, but they work in very different ways.
- **It is not magic or always right.** An AI system works out answers from the data and goals it was given. That is why it can be wrong.

Narrow AI and general AI

Almost every AI you use today is **narrow AI**. It is built for a specific task or a small set of tasks, such as a voice assistant that sets alarms and answers simple questions.

Artificial general intelligence (AGI) is a different idea. It would learn and use knowledge across a wide range of tasks, at the level of a person or better.

> **AGI does not exist today:** IBM states that this level of AI is theoretical and that no known system comes close to it. Treat claims that a product is "human-level in everything" with care.

In the next section you will see where the term came from. A group of researchers used the words "artificial intelligence" in a funding proposal in 1955, and the field has gone through big hopes and hard slumps since then.

A short history of AI

AI is not new. People have worked on it for more than seventy years. The story has big hopes, sharp disappointments and a few moments that changed everything.

Key moments

- **1950:** Alan Turing publishes the paper "Computing Machinery and Intelligence". It begins with a simple question: "Can machines think?"
- **1955–1956:** Four researchers, John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon, propose a two-month summer study at Dartmouth College. Their proposal uses the words "artificial intelligence". It rests on one idea: intelligence can be described so precisely that a machine can be made to simulate it.

- **1974–1980:** The first "AI winter". Funding and interest fall after a critical 1973 report in the UK, called the Lighthill report, and after the US agency DARPA cuts its support for academic AI research.
- **Mid-1980s:** Interest in neural networks, a way of building AI that is loosely inspired by the brain, returns.
- **From 1987:** A second slump begins. The market for special LISP computers collapses, and the AI systems of that time prove costly and fragile.
- **May 1997:** IBM's Deep Blue beats world chess champion Garry Kasparov 3.5 to 2.5 in a six-game match. It is the first computer to beat a reigning champion under standard tournament rules.
- **2012:** A system called AlexNet wins the ImageNet image-recognition contest. This helps start the machine-learning boom you see today.
- **2017:** Researchers publish "Attention Is All You Need". It introduces the Transformer, the design behind today's large language models.
- **30 November 2022:** OpenAI releases ChatGPT. It reaches one million users in five days and 100 million monthly users in about two months.

> ⓘ **A pattern to notice:** AI has moved in waves: high hopes, then disappointment and lower funding, then a new idea that restarts the climb. Knowing this helps you stay calm when a new claim sounds too big.

Why does this matter? Many of today's tools are built on ideas that are decades old. What changed is the amount of data and computing power available to run them.

AI, machine learning and deep learning

People use "AI", "machine learning" and "deep learning" as if they mean the same thing. They do not. They fit inside each other, like nested boxes.

Artificial intelligence (the biggest box)
Any technology that lets machines simulate human learning and related abilities.

Machine learning (inside AI)
A group of methods where algorithms learn patterns from data, instead of being given every rule.

Deep learning (inside machine learning)

Machine learning that uses neural networks with many layers. Today's chatbots and image generators are built this way.

How deep learning differs from classic machine learning

- **Features:** In classic machine learning, human experts choose which details of the data matter. Deep learning finds most of them on its own.
- **Type of data:** Classic machine learning usually works on neat, structured data. Deep learning can work on raw data such as text and images.
- **Labels:** Classic methods often need labelled examples. Deep learning does not always need them.
- **Amount of data:** Deep learning needs much more data to reach high accuracy.

> ☆ **Easy way to remember:** Every deep learning system is machine learning, and every machine learning system is AI. But not every AI system uses machine learning.

The next chapter looks inside the middle box. You will see how machines learn from data.

How Machines Learn

See how a machine can find patterns in data, and why the quality of that data decides everything.

Learning from data, not from rules

Normally, a programmer writes exact rules: if this happens, do that. This works well for tasks with clear rules. It fails for tasks where the rules are too many or too hard to write down, such as telling spam emails from real ones.

Machine learning takes a different path. Instead of writing the rules, you give the machine many examples. Its algorithms learn patterns from this **training data** and then make inferences about new data they have not seen before.

The key words

- **Features:** The facts about each example, turned into numbers. A house, for example, can be described by its area, number of bedrooms and age.
- **Model:** What the machine learns. It is a set of adjustable values, called parameters, tuned during training.
- **Inference:** Using the trained model on new data to get an output.
- **Generalisation:** The real goal. A model that does well on its training data should also do well in real use.

> ⓘ **Memorising is not learning:** A model can memorise its training data and then fail on anything new. This is called overfitting. Testing the model on data it has not seen helps catch this early.

Think of a student who memorises last year's question paper. They score well on that paper and badly on the real exam. A model that has learned the pattern behaves like a student who understood the subject.

Three ways to learn

Machines learn in three main ways. The difference is what kind of feedback they get.

Supervised learning: learning with an answer key

Each training example comes with the correct answer, called a label. The model learns to predict the right output. A spam filter trained on emails marked "spam" and "not spam" is a classic case. If the answer is a category, this is classification. If it is a number, such as a price, this is regression.

Unsupervised learning: finding patterns alone

There are no answers given. The model looks for structure in the data by itself. It can group similar items together, for example shoppers with similar habits. It can also find items that often go together, which is how many recommendation systems work.

Reinforcement learning: learning by trial and error

The model takes actions and gets a reward when it does well. Over time it learns which actions earn the most reward. This is used in robotics and in video games.

> ☆ **Which one is it?:** Ask what feedback the machine gets. A correct answer for every example means supervised. No answers at all means unsupervised. A reward or score after actions means reinforcement.

Why data quality matters

Google's machine learning course puts it simply: your model is only as good as the data it trains on. A good dataset helps the model reach its goal. A poor one holds it back.

Common problems with data

- **Missing values:** Some facts are blank.
- **Duplicates:** The same record appears many times.
- **Wrong values:** A number was typed or measured incorrectly.
- **Wrong labels:** A spam email marked "not spam", or the other way round.
- **Faulty sources:** One part of the data comes from a broken sensor or process.

Data must also fit the task. Google gives this example: when building a spam detector, you may keep messages sent by bots, because bots send spam. When studying how people

search, you would remove them.

When data is not fair

IBM explains that bias can enter an AI system through its training data. If the data is too small, incomplete or does not represent all the people it will be used on, the model learns that gap. This is called sample or selection bias.

> **Bad data means bad answers:** A clever model cannot fix poor data. Before you trust an AI result, ask: what data was this trained on, and who is missing from it?

How Modern AI Models Work

Look inside neural networks and large language models, and learn what they can and cannot do.

Neural networks in plain words

Most modern AI is built on **neural networks**. A neural network is a machine learning model made of many simple units called neurons, arranged in stacked layers.

Input layer

Takes in the raw facts, such as the words of an email or the pixels of a photo.

Hidden layers

Combine the inputs step by step into new, more useful patterns.

Output layer

Gives the final result, such as a number or a probability.

Every link between neurons has a **weight**. A weight works like a dial. It sets how strongly one input affects the result. Learning means turning these dials to the right positions.

How a network trains

- **Predict:** The network makes a guess for a training example.
- **Measure the error:** A loss function shows how far the guess is from the right answer.
- **Send the error backward:** This step is called backpropagation. It finds how much each weight added to the mistake.
- **Adjust slightly:** A method called gradient descent nudges the weights to reduce the error.
- **Repeat:** The cycle runs over the training data again and again.

> **What does 'deep' mean?:** Modern "deep" networks have many layers and millions or billions of parameters. This is where the name deep learning comes from.

What a large language model is

A **large language model (LLM)** is a deep learning model trained on huge amounts of text. It can understand and write human language. Chatbots like ChatGPT are built on LLMs.

How an LLM is built

- **Collect text.** Billions or even trillions of words from books, articles, websites and code. The text is cleaned to remove errors and duplicates.
- **Split into tokens.** Text is cut into small pieces called tokens. A token can be a word, part of a word, or a single character.
- **Learn to predict the next token.** IBM describes LLMs as "giant statistical prediction machines that repeatedly predict the next word in a sequence."
- **Fine-tune.** After the first round of training, the model can be refined so it follows instructions better and gives more helpful replies.

LLMs use a design called the **Transformer**. It was introduced in 2017 in the paper "Attention Is All You Need". It relies on a technique called attention, and it dropped the older recurrent and convolutional designs that came before it.

> **Key idea:** An LLM does not look up answers in a database. It writes text one token at a time, choosing what is likely to come next based on patterns it learned in training.

What these models do well, and where they fail

LLMs can write, summarise, translate and explain. They work fast and handle many tasks with one model. But they have clear limits that every user should know.

Hallucinations

Models can produce false or misleading information that sounds believable. Researchers Kalai and colleagues (2025) argue this persists because training and testing reward a confident guess more than an honest "I do not know".

Bias

Models can copy and even increase the biases found in their training data, which can lead to unfair or offensive output.

Cost

Training and running these models needs a lot of computing power, energy and money.

Their skills are also uneven. The Stanford AI Index 2026 describes "jagged" abilities. It notes that the top model read analog clocks correctly only 50.1% of the time, while success on a computer-use task test called OSWorld rose from 12% to about 66%. A model can be very strong at one task and weak at a simple one.

> **Do not judge by one good answer:** A model that answers a hard question well can still fail an easy one. Always check results that matter.

Using AI Wisely

Learn the main risks of AI, how to check its answers, and where the field is going next.

Bias, privacy and safety

Bias

IBM defines AI bias as biased results that come from human biases skewing the training data or the AI algorithm. It can enter in several ways:

- **Unrepresentative data:** The data is too small or leaves out groups of people.
- **Left-out data:** Important facts were never included.
- **Poor problem framing:** The goal given to the system was unclear or wrong.
- **Human labelling:** People who label data bring their own assumptions.

The effects can be serious. IBM points to harm for marginalised groups in hiring, policing and credit scoring.

Privacy

AI needs data, and some of that data is personal. IBM lists these privacy risks:

- Training data can include sensitive health, finance or biometric information.
- People may not know their data is being collected.
- Data shared for one purpose can end up training an AI for another.
- A model can accidentally leak one user's private details to others.

Safety

Safety problems are growing along with use. The Stanford AI Index 2026 counts 362 documented AI incidents in the latest year, up from 233 in 2024. In the US, the standards body NIST released its AI Risk Management Framework on 26 January 2023. It is a voluntary guide for managing AI risks to people, organisations and society.

Protect your own data: Do not type passwords, bank details, medical records or other private information into an AI tool unless you are sure how it is stored and used.

Checking AI answers

AI replies often sound sure and polished, even when they are wrong. As you saw in Chapter 3, models can write false statements that read like facts. The fix is a simple habit: check before you rely on it.

1. Check the numbers and names

Dates, amounts, names and quotes are the parts most often wrong. Confirm each one in a trusted source.

2. Ask for sources, then open them

If a reply names a source, open it. Make sure it exists and really says what the AI claims.

3. Cross-check with a second source

Compare the answer with an official site, a textbook or an expert. Do not rely on one AI reply alone.

4. Be extra careful with high-stakes topics

For health, money, law and safety, treat AI as a starting point and talk to a qualified person.

About this checklist: These four steps are practical advice for readers. They are not taken from a single source. They follow from the known problem that models can sound sure when they are wrong.

Where AI is heading

AI is spreading quickly. The Stanford AI Index 2026 reports that:

- 88% of organisations now use AI.
- Generative AI reached 53% of people in three years.
- AI agents, which carry out multi-step tasks on a computer, improved from about 12% to about 66% success on one standard test.

People disagree on what this means for jobs. In the same report, 73% of experts expect a positive effect on jobs, while only 23% of the public do.

Rules are arriving

Governments are writing rules. The European Union's AI Act came into force on 1 August 2024. It sorts AI by risk:

- **Unacceptable risk:** Banned, for example social scoring.
- **High risk:** The strictest rules apply.
- **Limited risk:** People must be told they are dealing with AI, such as a chatbot.
- **Minimal risk:** Not regulated.

> ☆ **Keep learning:** This field changes fast. Pick one or two trusted sources, such as the Stanford AI Index or official guidance from NIST, and read them once a year. Dates and figures in this book are as of October 2026.